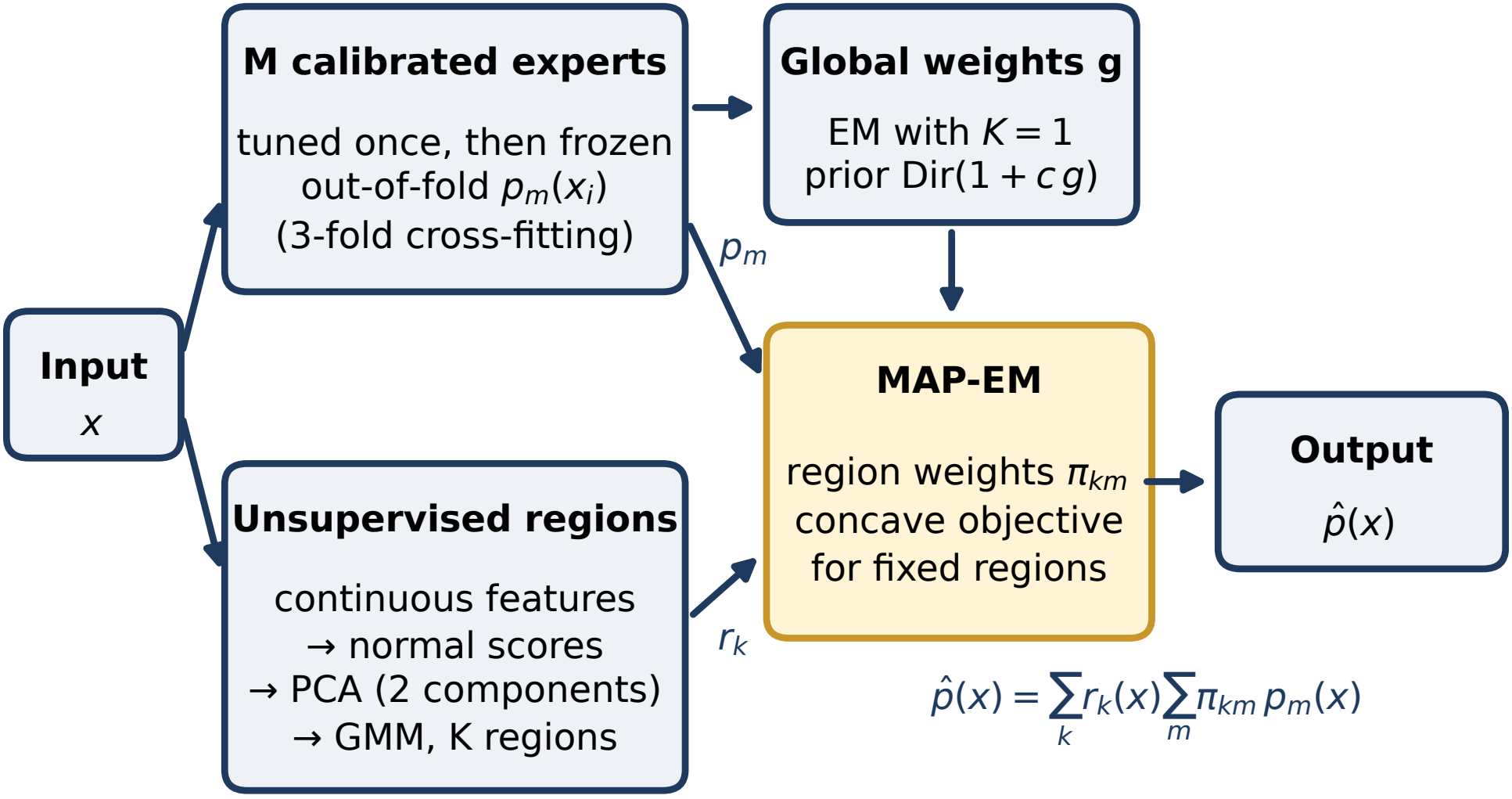
Input
x
M calibrated experts
tuned once, then frozen
out-of-fold $p_m(x_i)$
(3-fold cross-fitting)
Global weights g
EM with $K = 1$
prior Dir$(1 + c\,g)$
p_m
Unsupervised regions
continuous features
→ normal scores
→ PCA (2 components)
→ GMM, K regions
r_k
MAP-EM
region weights π_{km}
concave objective
for fixed regions
Output
$\hat{p}(x)$
$\hat{p}(x) = \sum_k r_k(x) \sum_m \pi_{km}\, p_m(x)$